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Assignment - 1

Section - A

Abstraction:-

1) • Hides complex details, shows essential features
• reduces system complexity.
• make design & development easier
• Enable modularity

2) levels of abstraction:-
Hardware : physical components (CPU, memory)
Machine : Binary instructions
Assembly : symbolic instruction (ADD, MOV)
High-level : C, Java, Python
Application : end-user program.

3) Basic functional units of computer:
• input - unit: takes input
• memory - unit: stores data / instructions
• ALU unit : performs calculations
• Control Unit : controls execution
• Output Unit: display result.

4) Differentiate
ALU: Arithmetic & logic operations
CU: controls & coordinations operations
Registers: smallest & fastest storage
Cache: stores frequently used data.

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5) Von Neumann Model

- CPU, I/O, memory connected
- same memory for data and instructions

Stored Program

instructions stored in memory executed sequentially.

6) Von Neumann bottleneck

Limited speed because CPU & memory share one path for data & instructions, so CPU often waits.

7) Moore's law

No. of transistors on a chip doubles over time, increasing performance (now slowing down).

8) IPC:-

- instruction executed per cycle
- Higher IPC - better performance.

9) Execution time = Total run time.

clock rate = cycle / sec

CPI = cycle per instructions.

IPC = 1/CPI

10) Bit: small unit of memory (0/1)

Byte : 8 bits

Signed: + and - values

unsigned: only + values

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11) Basic operations :-

- Arithmetic oper^n : +, −, /, *
- Logical oper^n : AND, OR, NOT
- Comparision : >, <, =
- Data movement : load / store

12) ASCII (7/8 bit)

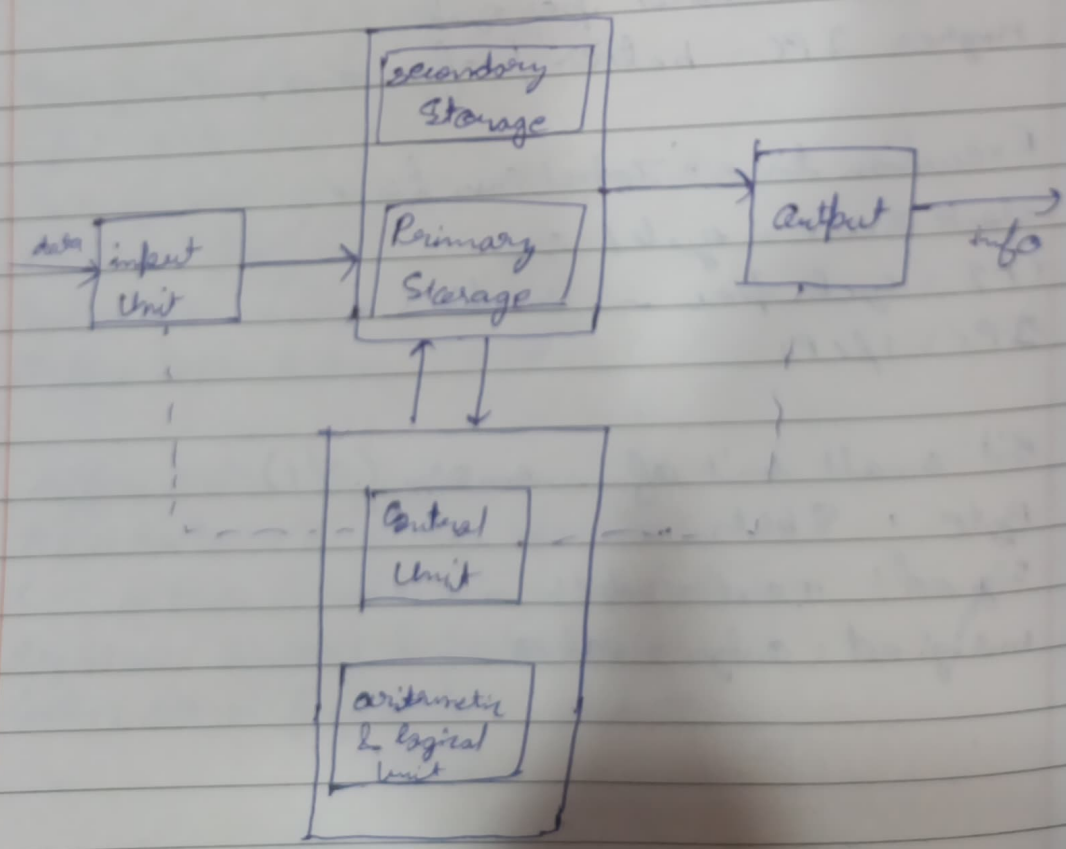
- Unicode (multi language)

Decimal

- Fixed point
- Floating point (IEEE 754)

Section-B

13)



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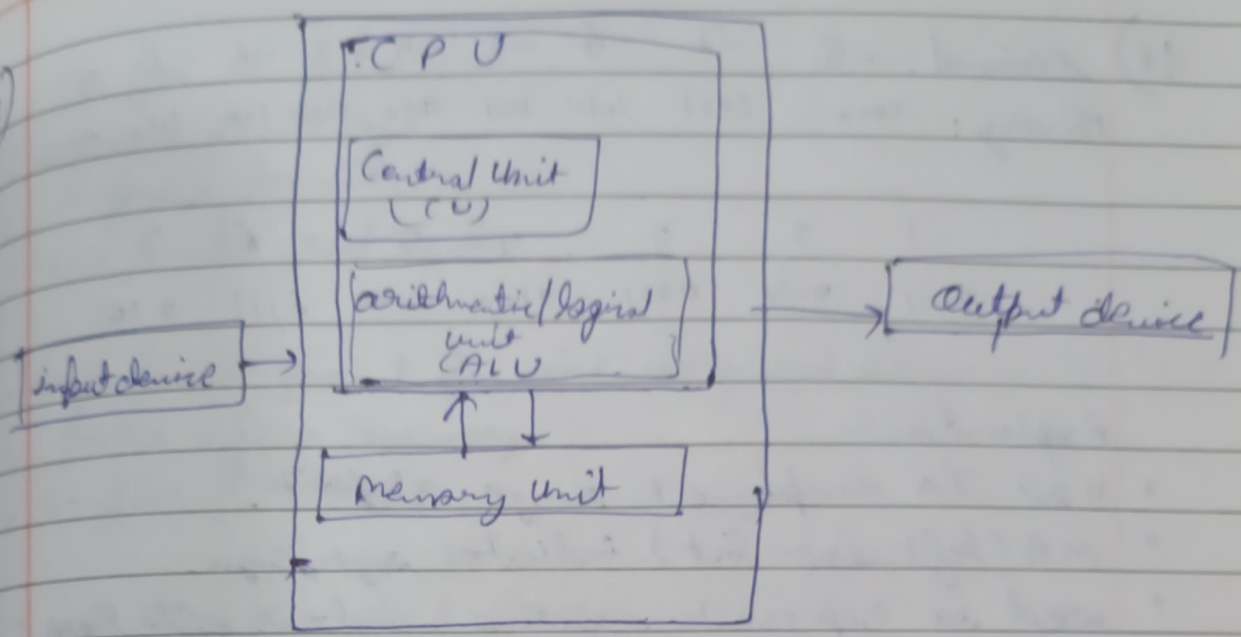
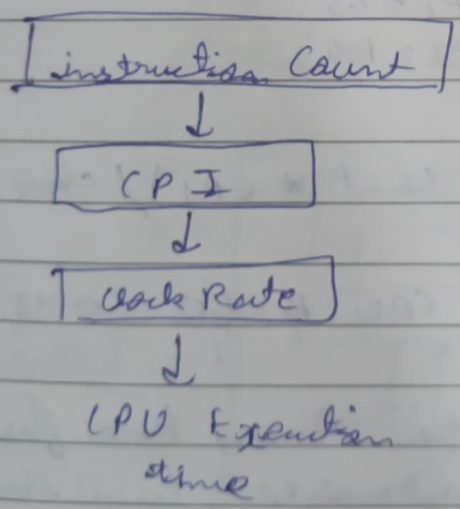


Stage memory stores both data & instructions. CPU computes with memory & I/O via common performed

15)



IPC = 1/CPI

Explanation :-

- CPU time depends on instruction count, CPI & clock rate
- Higher clock rate ↓ time ↑ CPI ↑ ⇒ time.
- IPC is inverse of CPI (IPC = better performance)

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16) Decimal: -8 -7 -6 -5 -4 -3 -2 -1 0
Binary: 1000 1001 1010 1011 1100 1101 1110 1111 0000

1 2 3 4 5 6 7
0001 0010 0011 0100 0011 0111 0101

Explanation:-
- 4 bit 2's complement range -8 to 7.
- MB (leftmost bit) indicates ~~sign~~ sign.
- used to represent negative numbers effective.

Section-C

17) CPU A: 3.9 Hz CPI = 1.5
CPU B: 2.2 9Hz CPI = 1.0

CPU Time = (instruction count * CPI) / clock rate

CPU A = $\frac{1.5}{3.0}$ = 0.5 CPU B = $\frac{1.0}{2.2}$ = 0.45

∴ CPU B has lower execution time, hence faster
↳ Reason lower CPI outweights lower clock rate

18) Effects of Cache MPSS
- Cache MPSS → CPU (stall cycles increase).
- IPC ↓ (fewer instructions per cycle)
- Execution time ↑

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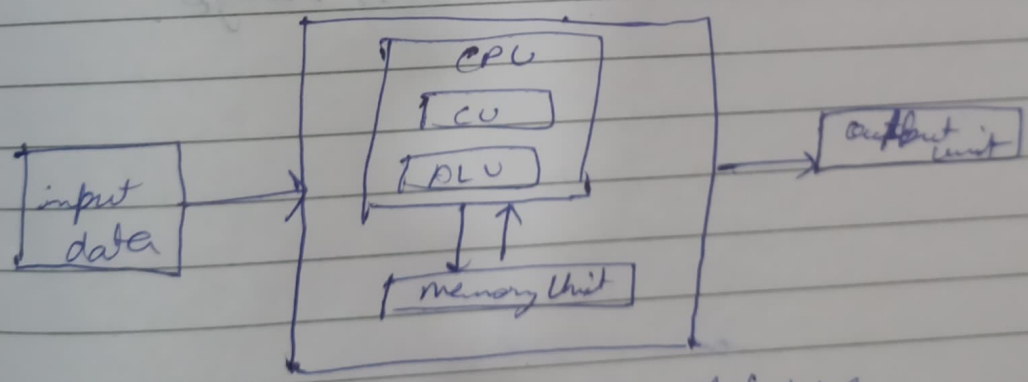
Fix for it
- increase cache size
- improve locality
- use multi level cache

- use fixed point for given language range
- faster, less memory
- simpler hardware

- not directly about performance
- limited by heat & power.
- modern gains via parallelism
- doesnot guarantee performance doubling.

Section-D

2)



Von-neuman architecture

- Design decision : single memory for design & decision
- Explanation:-
  - simplifies system design & reduces hardware cost.
  - But leads to Von-Neumann bottle neck due to char. bus.

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22) Performance:-

$$\text{CPU Time} = \frac{(\text{instruction} \times \text{CPI})}{\text{clock rate}}$$

23) Binary addition :-

+5 = 0101

+4 = 0100

result = 1001

interpretation: 1001 = -7 (2's complement)

Overflow

(+ve) + (+ve) → (-ve) ⟹ overflow

occurs when result exceeds range.